Horizon Bank

Customer Churn at Horizon Bank

"A Data-Driven Approach to Customer Retention"

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"Customer churn is not merely a performance metric for Horizon Bank, it reflects how customers experience services, engage with products, and ultimately decide whether to remain with the bank."

Horizon Bank

Executive Summary

⚠️ Problem

Horizon Bank experiences a relatively high customer churn rate of 20.4%, with substantial variation across countries and customer engagement levels. These differences indicate that churn is not uniformly distributed and may be driven by distinct customer characteristics and behaviors.

💡 Key Insights

- The overall churn rate is **20.4%** across **10,000 customers**
- **Germany shows the highest churn rate at 32.4%**, compared to **France (16.2%)** and **Spain (16.7%)**
- **I**nactive customers exhibit a churn rate of **26.9%**, nearly **double** that of active customers (**14.3%**)
- Churned customers hold a **higher average balance (€91,109)** than retained customers (**€72,745**), suggesting that churn is not primarily driven by customers' financial capacity, but rather by engagement and behavioral factors.

🎯 Objective

This study aims to identify the key drivers of customer churn at Horizon Bank by analyzing customer data across three root causes customer profile, financial behavior, and engagement & product usage based on customers from France, Germany, and Spain.

📌 Recommendations

Based on the findings, Horizon Bank should prioritize:

- Geo-specific retention strategies, particularly targeting high-churn markets such as Germany
- Engagement-focused initiatives, including reactivation programs for inactive customers
- Product and onboarding optimization, to improve customer engagement and reduce early-stage churn

Project Background

Customer churn poses a direct threat to long-term profitability and customer lifetime value in the banking industry.

With a churn rate of **20.4%**, Horizon Bank faces a substantial risk of revenue erosion if **high risk** customer segments are not addressed. Retaining existing customers is widely recognized as more cost effective than acquiring new ones, making churn analysis a strategic priority.

This study focuses on identifying the key drivers of churn, rather than solely predicting churn, in order to support data-driven retention strategies. To support **data informed retention strategies**.

Source : *Banking Customer Churn Analysis : Understanding Customer Behavior Through Churn Analysis|BillingPlatform*

⚠️ Problem Statemen

Despite a stable customer base, Horizon Bank is experiencing a notable level of customer attrition. The bank lacks clear visibility into where, who, and why customers are churning, making it difficult to design targeted retention strategies.

Project Objective

This analysis aims to identify the key factors associated with customer churn by examining customer profiles, financial behavior, and engagement patterns.
The findings are intended to support data-driven customer retention strategies.

KPI Overview

These KPIs provide a high level overview of customer base size and overall churn performance.

20,4%
Total Churn Rate

10,000
Total Customers

Germany
32,4% Hight Country Churn Rate

50-59 Age
56% Churn Rate

26,9%
Inactive Customers Churrate

💡 The KPI overview shows that churn is not evenly distributed across customers, but concentrated in specific segments particularly in Germany, customers aged 50–59, and inactive customers indicating clear patterns that require deeper root cause analysis.

Source : *DEEP Mileston 3 Visualization by Tableau*

Approach to Churn Analysis

Horizon Bank

1

Defining the Problem

Clarify the business context and define churn related questions that are relevant to customer retention.

2

Clean and prepare historical customer data, including handling irrelevant fields and ensuring data consistency.

Data Preparation

3

Data Analysis

Explore customer characteristics and behavioral patterns using descriptive analysis to identify factors associated with churn.

4

Conduct exploratory data analysis to identify patterns and factors associated with churn.

Data Visualization

5

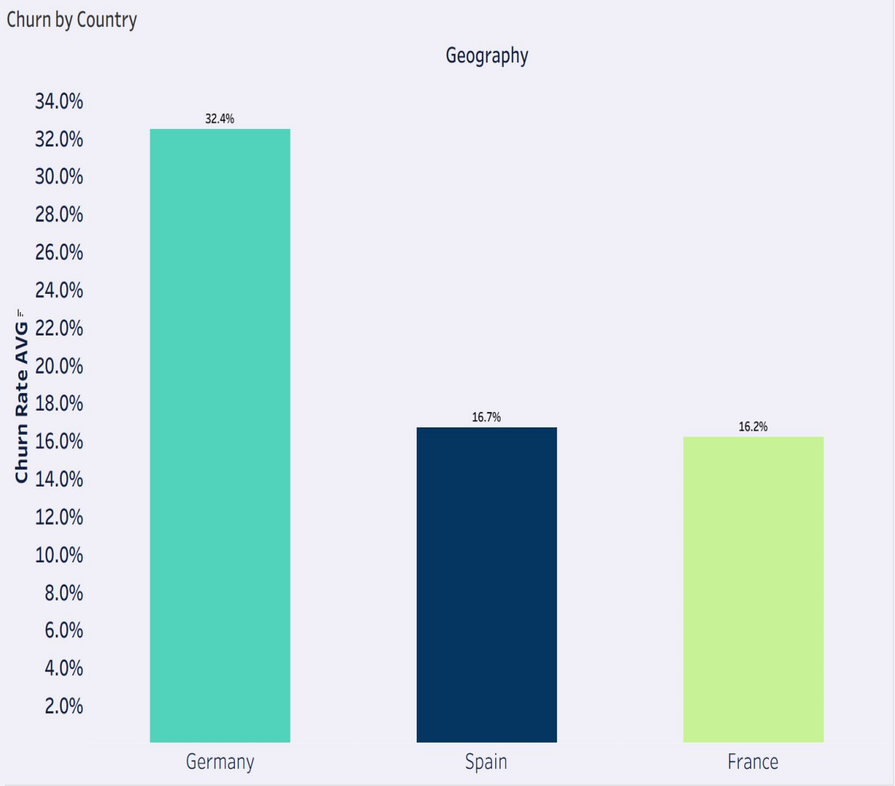
Recommendations

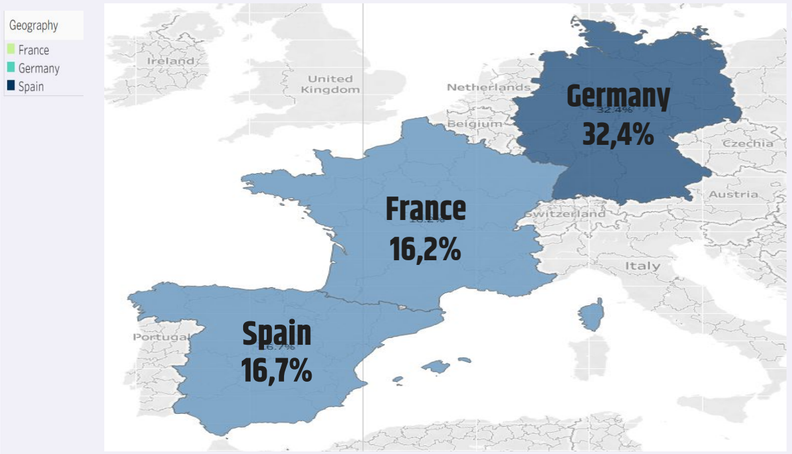
Translate analytical findings into practical, data-driven recommendations to support sustainable customer retention strategies.

Horizon Bank

“The following section examines customer characteristics to identify which segments are most associated with churn.”

Customer Profile — Geography





Customer distribution across France, Germany, and Spain.

Tabel Country

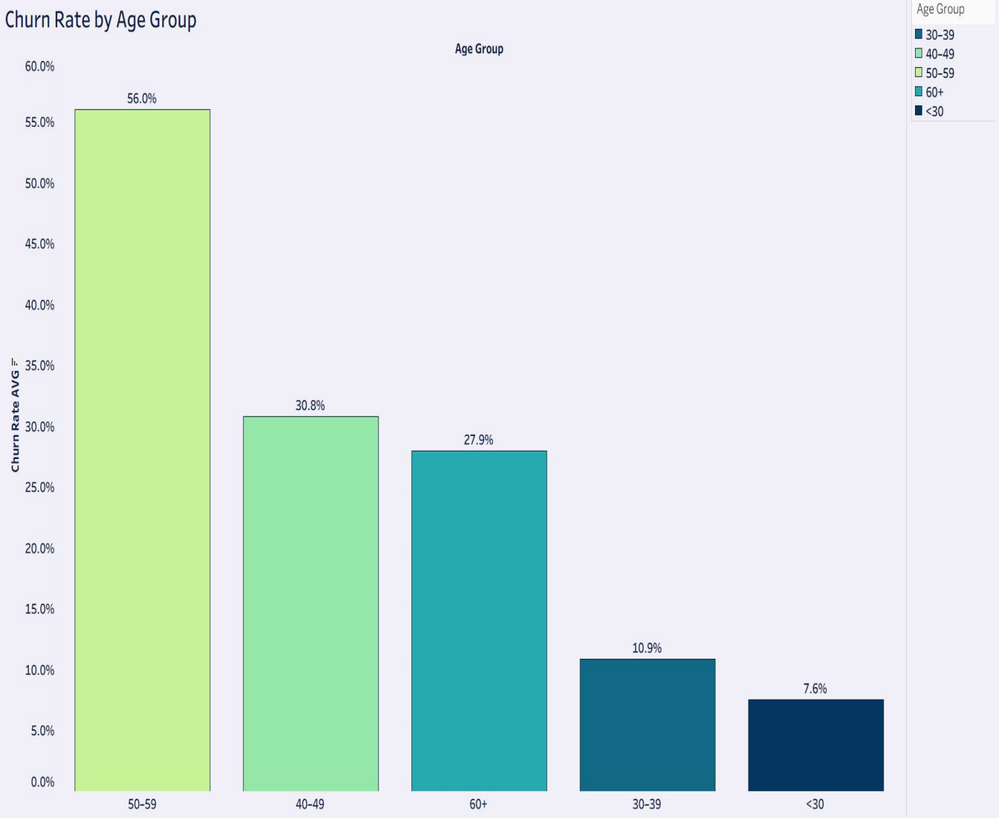
Geogr..	Churn Rate AVG	Count of Customer Id
Germany	32.4%	2,509
Spain	16.7%	2,477
France	16.2%	5,014

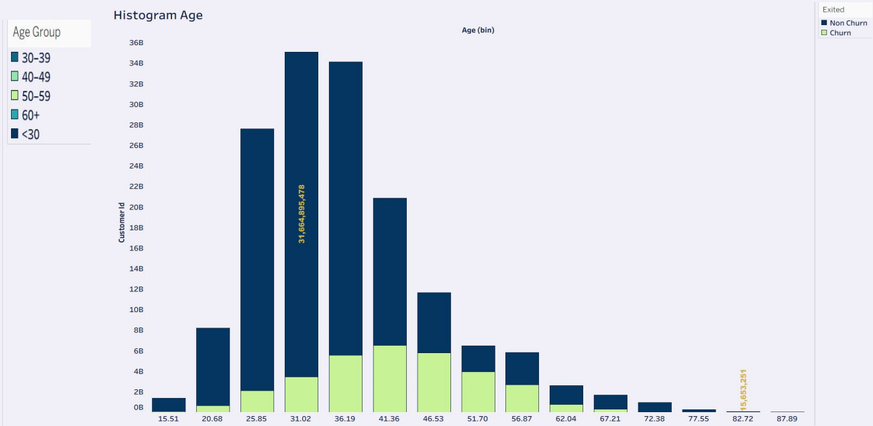
Germany records the **highest** churn rate at **32.4%**, compared to 16.7% in Spain and 16.2% in France.

🧭 This indicates that churn risk varies significantly by geography, suggesting the need for country specific retention strategies.

Source : *DEEP Mileston 3 Visualization by Tableau*

Customer Profile – Age





Tabel Age Group

Age Group	Churn Rate AVG	Customer Id
50-59	51.6%	5,994,418,583
40-49	25.5%	19,489,134,290
60+	24.8%	4,299,259,564
30-39	8.1%	35,303,606,116
<30	5.2%	13,586,501,117

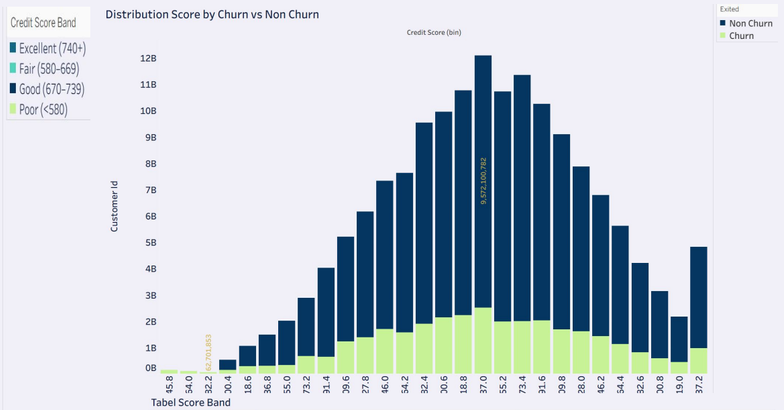
stomers aged **50–59** show the **highest** churn rate at **56.0%**, followed by 40–49 (30.8%) and 60+ (27.9%),

🧭 while younger segments 30–39 (10.9%) and <30 (7.6%) show much lower churn.

Churn risk increases sharply with age, suggesting that older customers may experience **lower engagement** or **unmet financial needs**, increasing their likelihood of attrition..

Source : *DEEP Mileston 3 Visualization by Tableau*

Customer Profile – Credit Score



Tabel Score Band

Credit Score Band	Churn Rate AVG	Customer Id
Excellent (740+)	20.2%	29,484,940,035
Fair (580–669)	20.6%	52,266,674,058
Good (670–739)	18.6%	38,097,355,579
Poor (<580)	22.0%	37,060,436,022

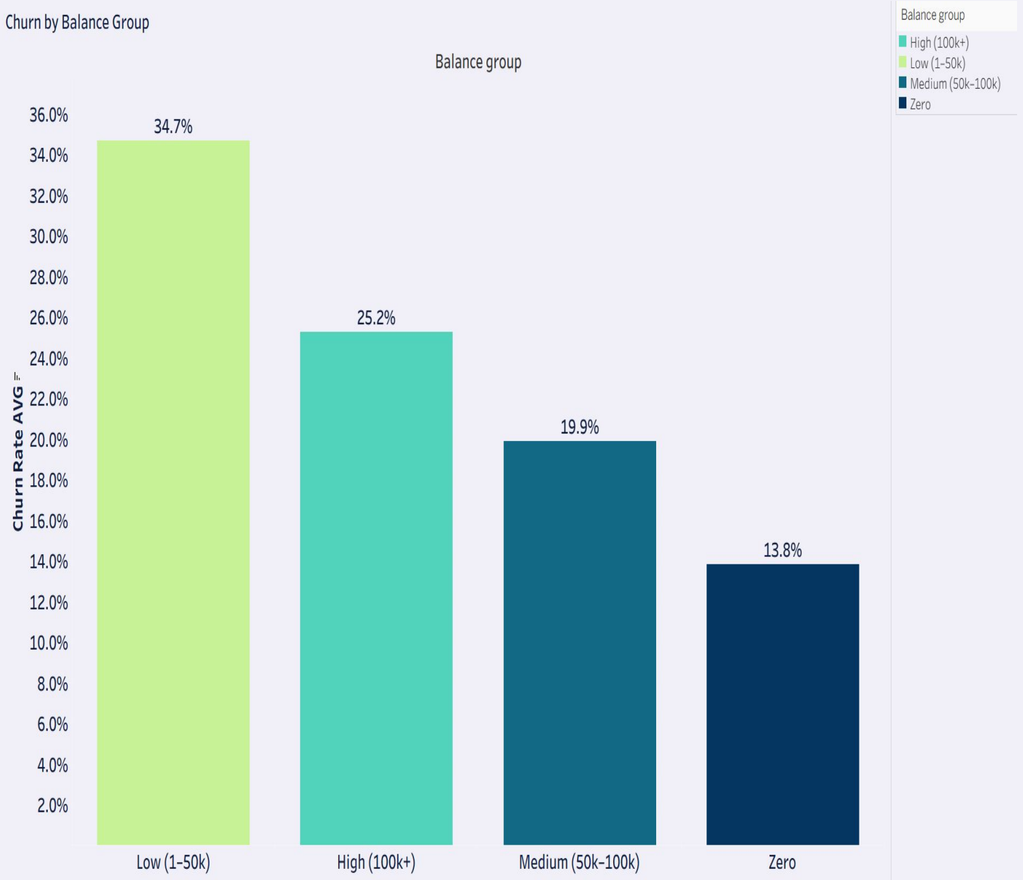
💡 Customers with **lower credit scores** exhibit higher churn rates compared to those with **stronger credit profiles**.

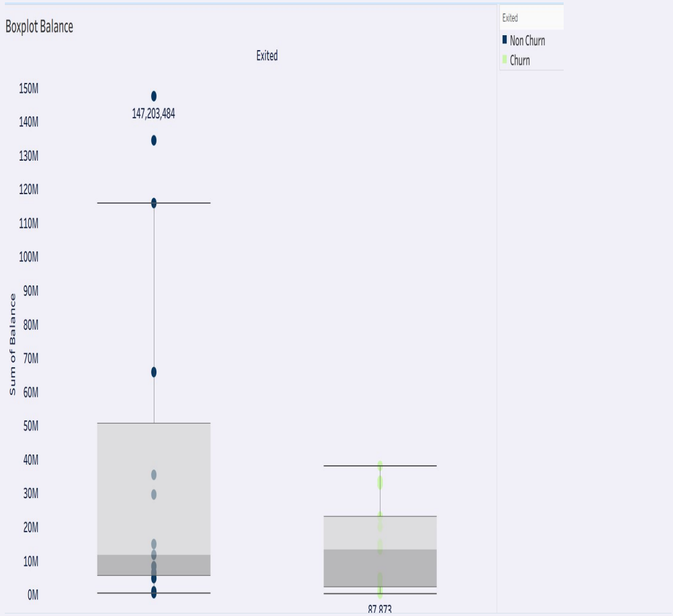
🧭 This suggests that **financial vulnerability** and **creditworthiness** are **significant** predictors of **customer attrition.**

Source : *DEEP Mileston 3 Visualization by Tableau*

“Customer profile characteristics help explain which customers are more likely to churn. The next section examines how financial behavior further influences customer attrition.”

Financial Behavior — Account Balance





Tabel Balance Group

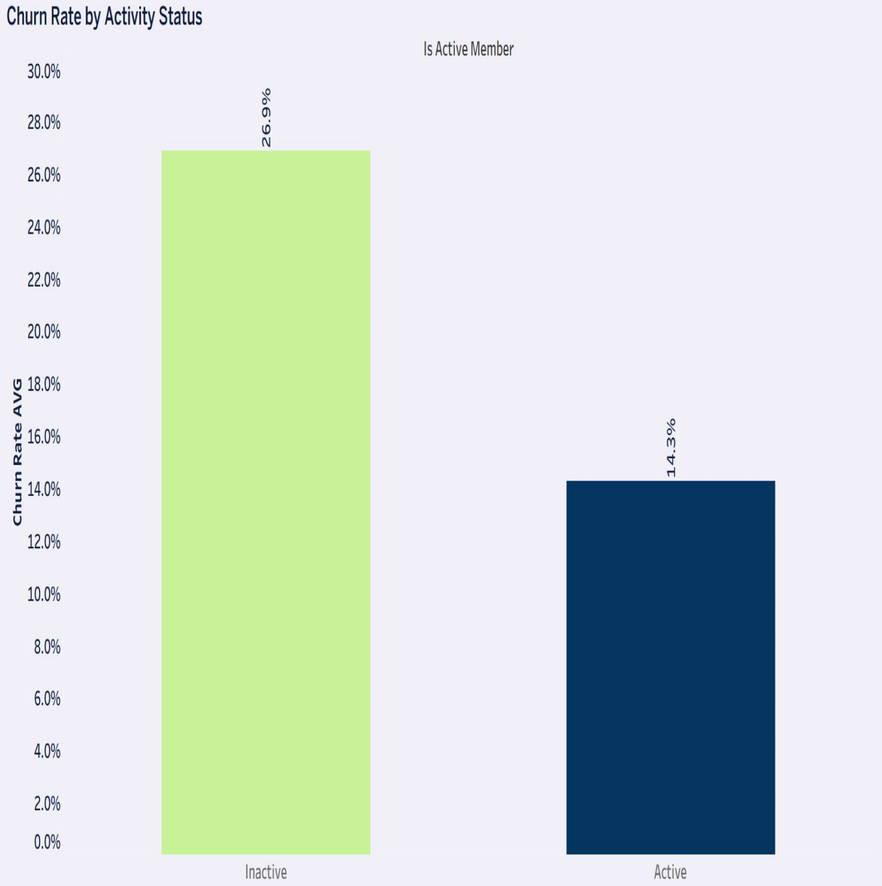
Balance group	Churn Rate AVG	Customer Id
Low (1-50k)	48.8%	642,684,951
High (100k+)	18.0%	29,937,030,638
Medium (50k-100k)	17.0%	10,151,053,288
Zero	13.9%	37,942,150,793

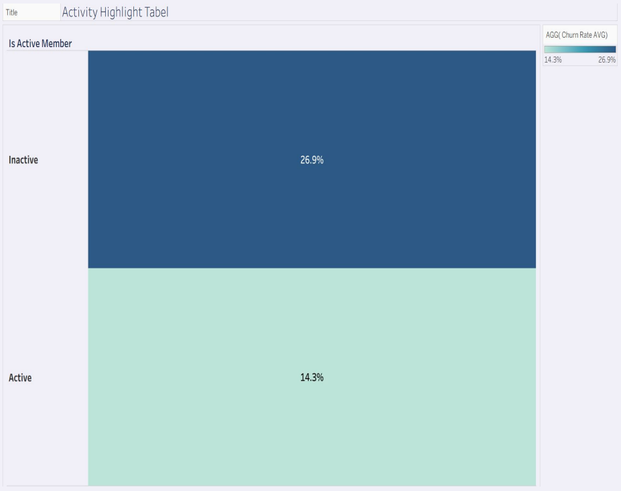
💡 Customers with low balances (1–50k) show the **highest** churn rate **(34.7%)**, while churn declines as account balance increases.

🧭 **Low balance** customers represent the **most at risk** segment and should be prioritized for retention and engagement initiatives.

Source : *DEEP Mileston 3 Visualization by Tableau*

Engagement & Product Usage — Activity Status





Tabel Active Group

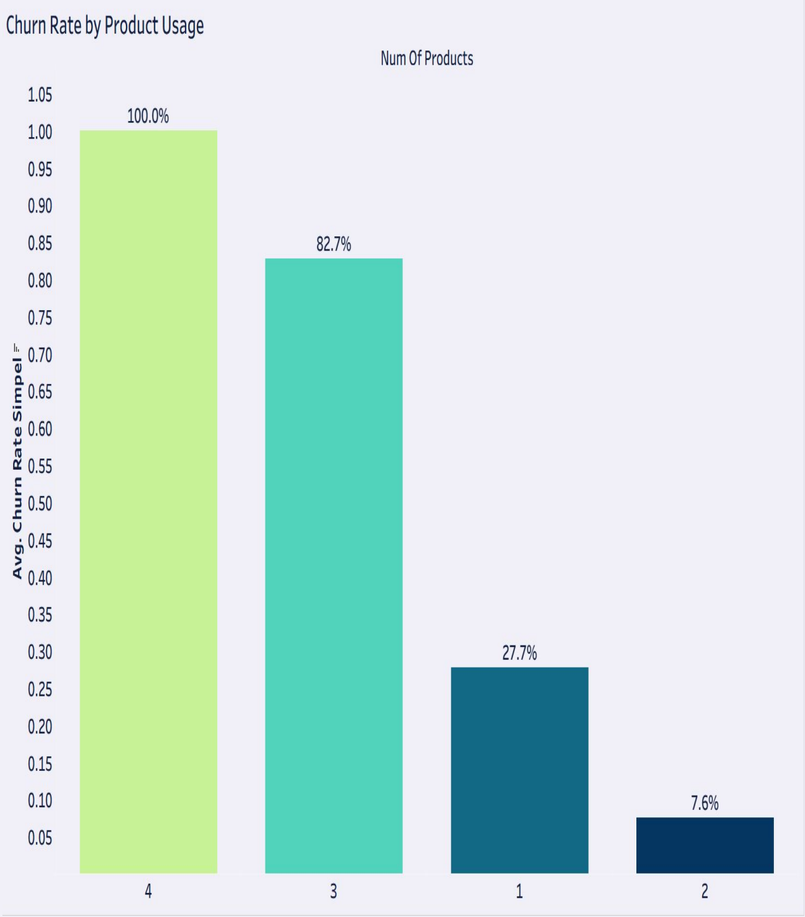
Is Active Member	Churn Rate AVG	Customer Id
Inactive	21.1%	38,020,850,001
Active	11.5%	40,652,069,669

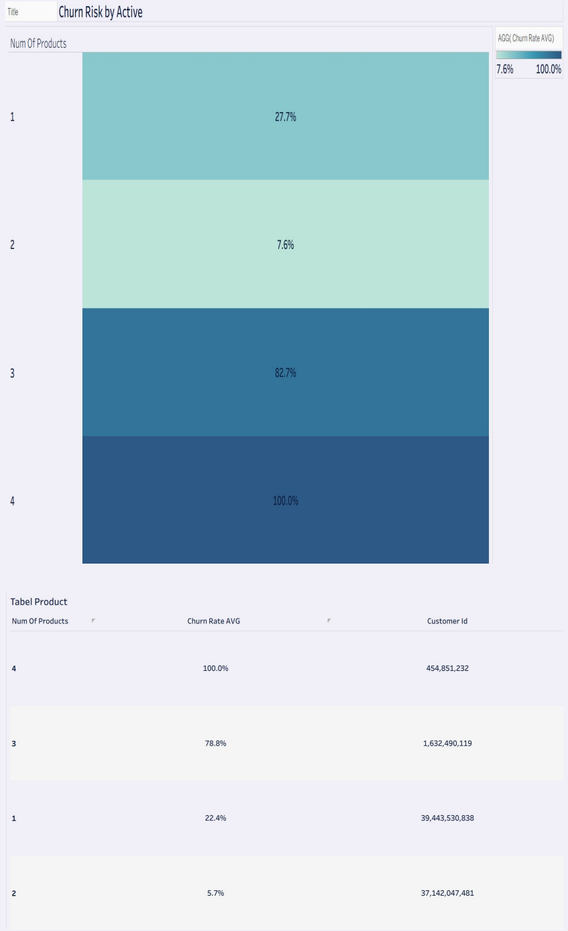
💡 Inactive members have a churn rate of 26.9%, nearly double that of active members at 14.3%.

🧭 Improving customer activity could potentially reduce churn risk by **over 12 percentage points**, highlighting customer engagement as a critical driver of retention.

Source : *DEEP Mileston 3 Visualization by Tableau*

Engagement & Product Usage — Product Adoption





While **multi product** adoption generally correlates with **lower churn** (notably at two products), **extremely high churn** observed at **higher product counts suggests potential overextension, misaligned product fit, or service complexity.**
This **indicates** that product quantity alone does **not guarantee retention**, and **highlights the importance of product relevance and customer experience quality**.

Customers with **one product** show a churn rate of **27.7%**, significantly higher than those with two products (7.6%). Churn rates increase sharply for customers holding three (82.7%) and four products (100%), **indicating** a non linear relationship between **product count** and **churn behavior**.

Source : *DEEP Mileston 3 Visualization by Tableau*

Key Drivers of Customer Churn

Root Cause	Evidence from RCA	Churn Risk	Business Meaning
Geography	Germany shows 32.4% churn vs ~16% in France and Spain	Country-level risk	Requires country-specific retention strategy
Age Group	Customers aged 50–59 show 56% churn rate	Aging segment risk	Older customers are less engaged with current services
Credit Score	Low credit score band has the highest churn rate	Financial risk profile	Customers with poor credit profiles are more likely to leave
Account Balance	Low balance customers show 34.7% churn rate	Low-value customer attrition	Customers with small balances require targeted engagement
Activity Status	Inactive customers: 26.9% vs Active: 14.3%	Engagement issue	Customer activity is a key retention driver
Number of Products	Customers with 1 product: 27.7% vs 2 products: 7.6%	Low product adoption	Cross-selling products significantly reduces churn

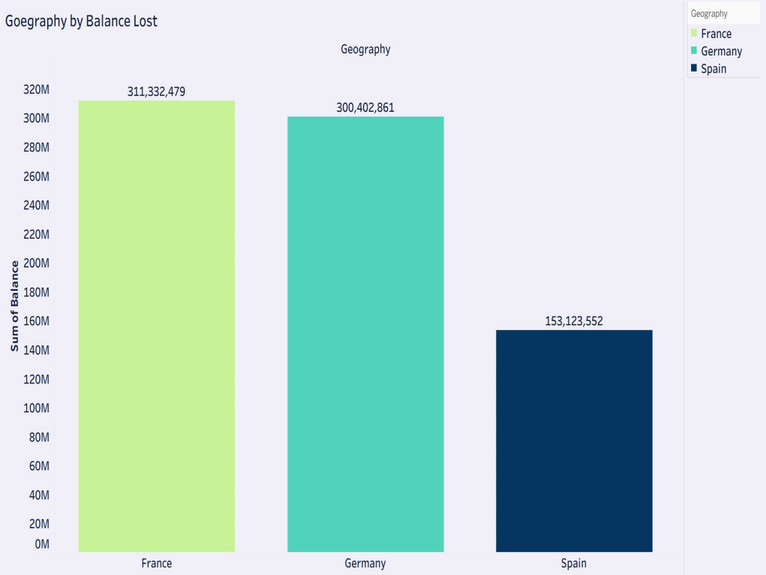
💡 Customer churn is primarily driven by geography (Germany), older age segments, low credit scores, low account balances, inactive status, and low product adoption.

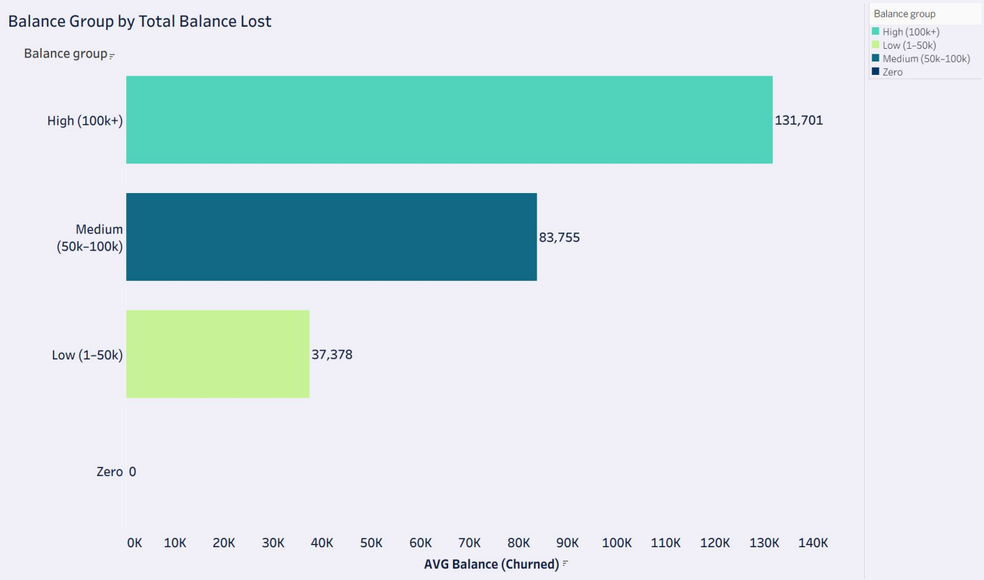
Business Impact of Customer Churn

2,037
Total Churned Customers

€185,588K
Total Balance Lost

€91,109
AVG Balance per Churned Customer





💡 Customer churn not only reduces the customer base but also results in substantial financial losses, as churned customers often hold significant account balances.

Source : *DEEP Mileston 3 Visualization by Tableau*

Data-Driven Retention Strategy

Root Cause Identified	Risk Segment	Recommended Action	Expected Outcome
Geography (Germany)	Customers in Germany	Country-specific retention campaign	Reduce high churn concentration
Age 50–59	Older customers	Personalized senior-friendly banking services	Increase engagement among aging segment
Low Credit Score	Financially risky profile	Offer financial advisory & credit improvement programs	Improve customer stability
Low Account Balance	Low balance customers	Targeted engagement & incentive programs	Prevent low-value customer attrition
Inactive Customers	Low activity segment	Re-engagement campaign & digital reminders	Increase customer activity
1 Product Users	Low product adoption	Cross-selling additional banking products	Increase stickiness & reduce churn

💡 By targeting the identified root causes with focused retention strategies, the bank can proactively reduce churn and improve long-term customer value.

Source : *DEEP Mileston 3 Visualization by Tableau*

Customer Churn at Horizon Bank

A Data-Driven Approach to Customer Retention

CLICK HERE

KPI Total Customers

10,000

Total Customers

KPI Churn Rate

20.4%

Total Churn Rate

KPI Churn Rate of Inactive Customers

26.85%

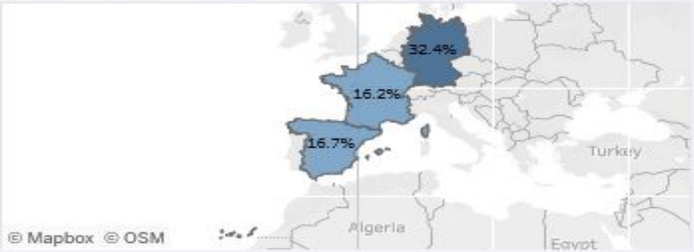
Active Customer Rate

KPI Active Rate

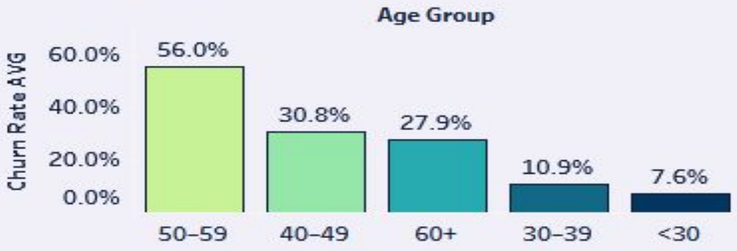
51.5%

Active Customer Rate

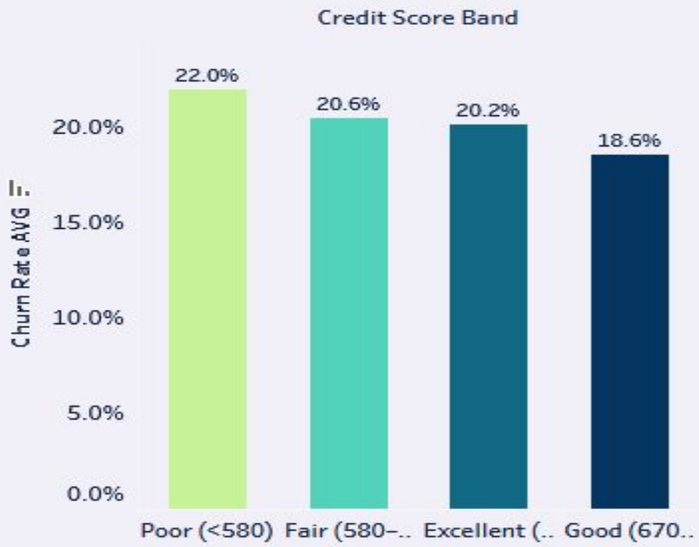
MAP Churn by Geography



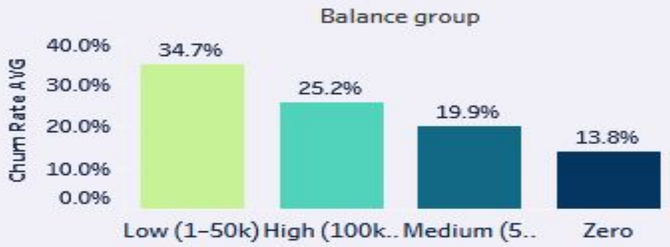
Churn Rate by Age Group



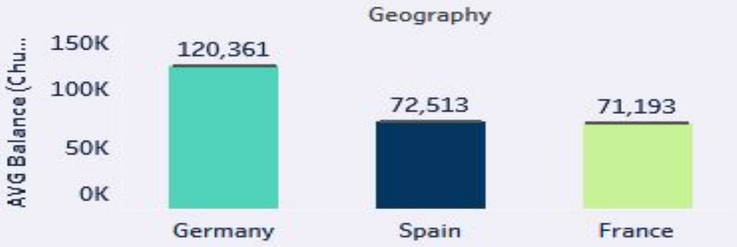
Churn by Credit Score Band



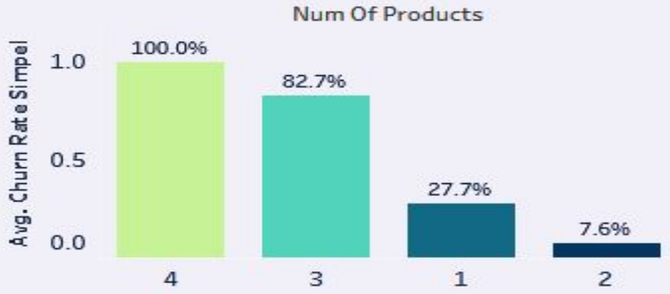
Churn by Balance Group



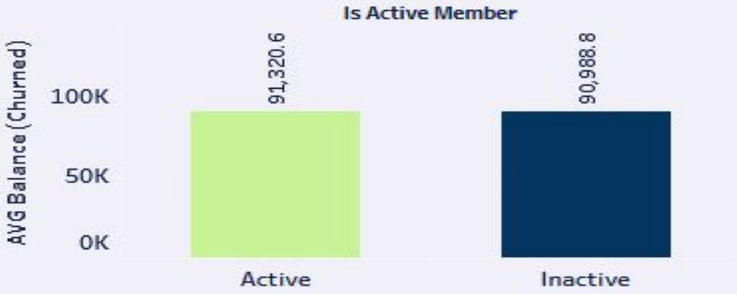
Account Balance by Geography



Churn Rate by Product Usage



Account Balance by Activity Status



Customer Distribution by Number of Products

Source : *DEEP Mileston 3 Visualization by Tableau*

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Horizon Bank

APPENDIX

Horizon Bank

Dataset Overview

Dataset Overview
The analysis is based on a snapshot of Horizon Bank's customer dataset containing **10,000 customers** across three countries.

Key Dataset Characteristics

- Countries: **France, Germany, Spain**
- Total customers: **10,000**
- Churn indicator: **Exited (1 = churned, 0 = retained)**

Key Variables Used

- Customer Profile: Geography, Age, Credit Score
- Financial Behavior: Balance, Estimated Salary
- Engagement & Product Usage: Tenure, IsActiveMember, Number of Products

Dataset scema, columns summary and data type.

Column Name	Description	Type
CustomerId	Unique customer ID	Numeric
Geography	Customer country	Categorical
Age	Customer age	Numeric
Balance	Account balance	Numeric
NumOfProducts	Number of products	Numeric
IsActiveMember	Activity status	Binary
Exited	Churn flag	Binary

Source : *DEEP Mileston 3 Visualization by Tableau*

Stakeholder Identification

D
DECISION MAKER

Head of Retail Banking

Decide/ approve churn reduction strategies

A
ACCOUNTABLE

Customer Retention Manager

Ensure project delivers actionable strategies

R
RESPONSIBLE

Data Analyst/ Data Scient

Conduct churn analysis. build predictive model, generate insights

C
CONSULTED

Marketing and Product Teams

Provide input on retention campaign and current condition regarding customer

I
INFORMED

Senior Management

Update on new strategies and expected to implement change

Project Scope

ANALYZE

Overall churn distribution (**20.4%** total churn rate). Churn patterns across **France, Germany, and Spain.**

IDENTIFY

High-risk customer segments based on:

- Geography
- Engagement level
- Product usage

DEVELOP

A KPI framework to summarize churn risk at the root-cause level.

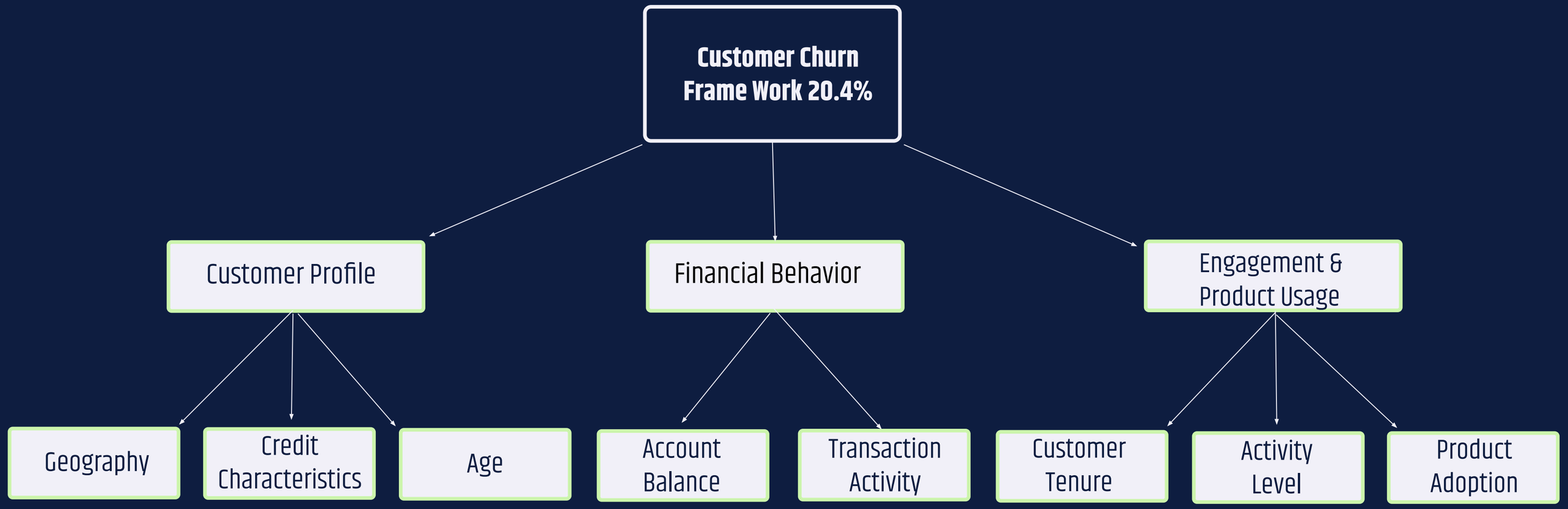
PROPOSE

Data-driven retention insights aligned with observed churn patterns. *This analysis focuses on understanding churn drivers, not predicting churn.*

Boundaries

- No predictive or machine learning models are developed
- No causal inference is performed; relationships are correlational
- The dataset represents a historical snapshot of customer behavior
- Transaction activity is approximated using Estimated Salary as a proxy
- Findings are specific to the analyzed dataset and period

Roote Cause Analysis



Cleaning Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   RowNumber        10000 non-null  int64
 1   CustomerId       10000 non-null  int64
 2   Surname          10000 non-null  object
 3   CreditScore      10000 non-null  int64
 4   Geography        10000 non-null  object
 5   Gender           10000 non-null  object
 6   Age              10000 non-null  int64
 7   Tenure           10000 non-null  int64
 8   Balance          10000 non-null  float64
 9   NumOfProducts    10000 non-null  int64
 10  HasCrCard        10000 non-null  int64
 11  IsActiveMember   10000 non-null  int64
 12  EstimatedSalary  10000 non-null  float64
 13  Exited           10000 non-null  int64
dtypes: float64(2), int64(9), object(3)
memory usage: 1.1+ MB
```

```
#Check duplikat data
df.duplicated().sum()
```

```
np.int64(0)
```

```
#Untuk churn analysis, ID & Surname tidak dipakai untuk analisis
df = df.drop(columns=["RowNumber", "CustomerId", "Surname"])
```

- The dataset is complete and requires only minimal data cleaning.
- representing **10,000 customers**, **14 columns**, and data types : **int64 (9 Columns), float64(2 columns), object(3 columns)**.
- No duplicate removal is necessary.
- Rationale for removing these columns : **RowNumber (sequential index with no analytical value), CustomerId (Unique identifier, not predictive churn), Surname (Personal Identifier, not relevan for churn analysis)**.
- After removing columns we use **11 columns** The dataset is now clean, well-structured, and suitable for EDA, Feature encoding, Machine learning modeling, Visualization and dashboard creation (e.g., Tableau)

```
#cek data setelah cleaning
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 11 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CreditScore      10000 non-null  int64
 1   Geography        10000 non-null  object
 2   Gender           10000 non-null  object
 3   Age              10000 non-null  int64
 4   Tenure           10000 non-null  int64
 5   Balance          10000 non-null  float64
 6   NumOfProducts    10000 non-null  int64
 7   HasCrCard        10000 non-null  int64
 8   IsActiveMember   10000 non-null  int64
 9   EstimatedSalary  10000 non-null  float64
 10  Exited           10000 non-null  int64
dtypes: float64(2), int64(7), object(2)
memory usage: 859.5+ KB
```

Source : *DEEP Mileston 2 EDA Python*

Data Binning

```
#data bins pengelompokan score brand berdasarkan labels
df['Credit_Band'] = pd.cut(
    df['CreditScore'],
    bins=[300, 580, 670, 740, 800, 900],
    labels=['Poor', 'Fair', 'Good', 'Very Good', 'Excellent']
)
```

```
#bin pengemlompokan dan pelabelan
df['Tenure_Group'] = pd.cut(
    df['Tenure'],
    bins=[0,2,5,10],
    labels=['0-2 yrs','3-5 yrs','6-10 yrs']
)
```

```
#churn vs geography
sns.histplot(
    data=df,
    x="Balance",
    hue="Exited",
    bins=30,
    palette=["#0E1D40", "#CCF5AC"],
    kde=True,
    alpha=0.6
)
```

To improve interpretability and align the analysis with business segmentation, several continuous variables were transformed into categorical groups.

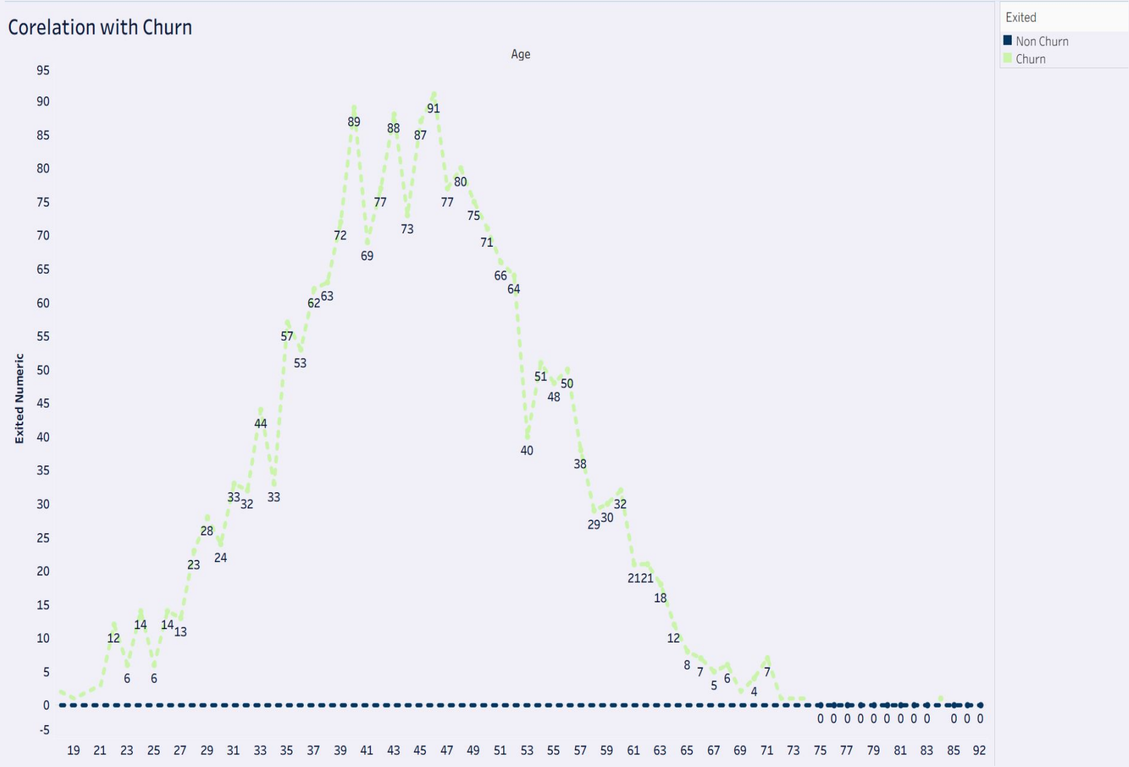
Purpose of binning:

- Simplify interpretation
- Reduce noise from extreme values
- Enable clearer comparison across customer segments

Binning Applied

- Age ⟶ Age Group
- Credit Score ⟶ Credit Band
- Tenure ⟶ Tenure Group

Correlation with Churn Age



📊 Correlation with Churn — Age

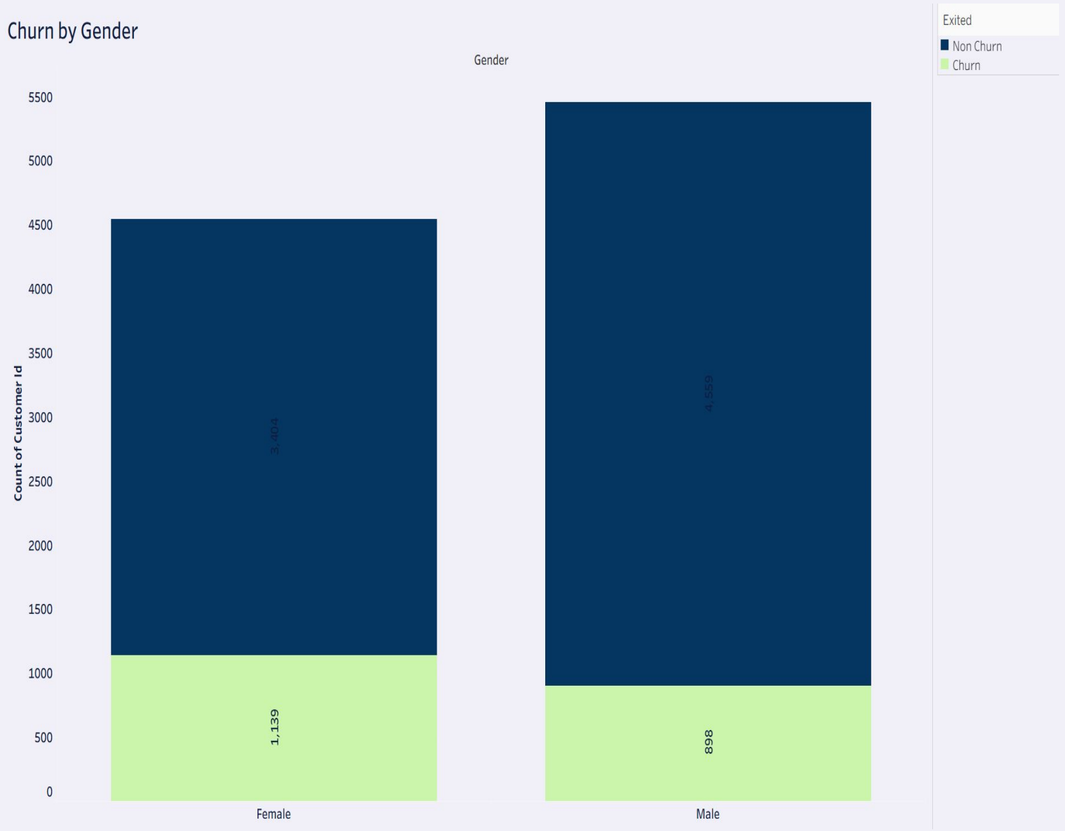
Insight

Churn terkonsentrasi di usia 35–55, puncak sekitar 45–50.

Business Meaning

Nasabah usia produktif paruh baya adalah segmen risiko utama yang perlu prioritas retensi.

Churn Rate by Gender



📊 Churn by Gender

Insight

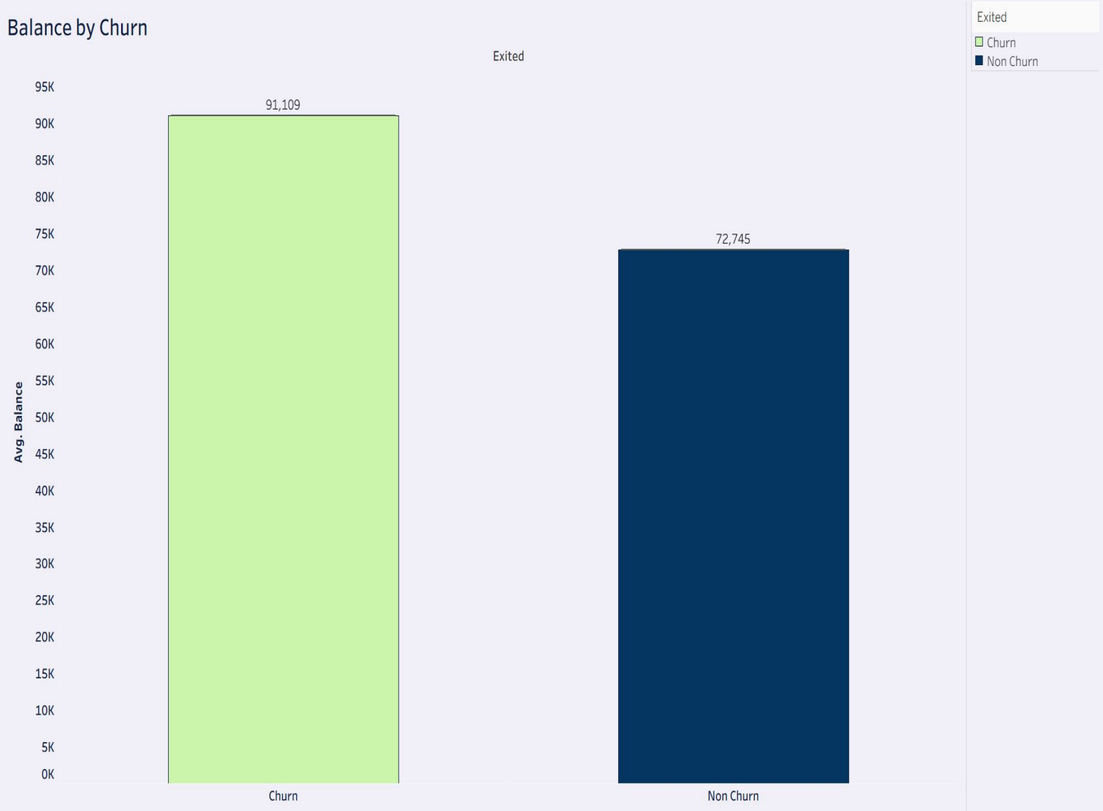
Churn Female: 1.139 (25%) lebih tinggi dari Male: 898 (19.7%).

Business Meaning

Nasabah wanita menunjukkan risiko churn lebih tinggi → perlu pendekatan engagement berbeda.

Source : *DEEP Mileston 3 Visualization by Tableau*

Churn by Balance



📊 Balance by Churn

Insight

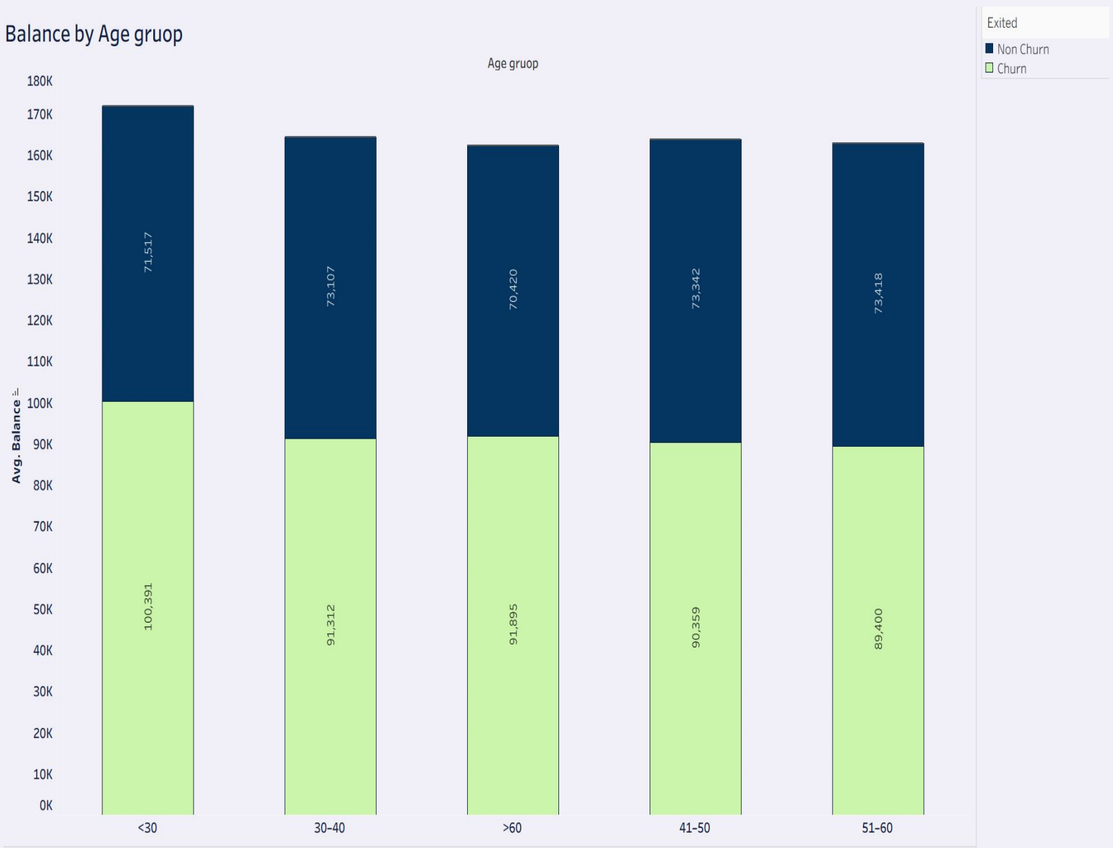
Rata-rata saldo churn: 91,109 lebih tinggi dari non-churn: 72,745.

Business Meaning

Bank kehilangan nasabah bernilai tinggi saat churn terjadi.

Source : *DEEP Mileston 3 Visualization by Tableau*

Balance by Age Group



📊 Balance by Age Group

Insight

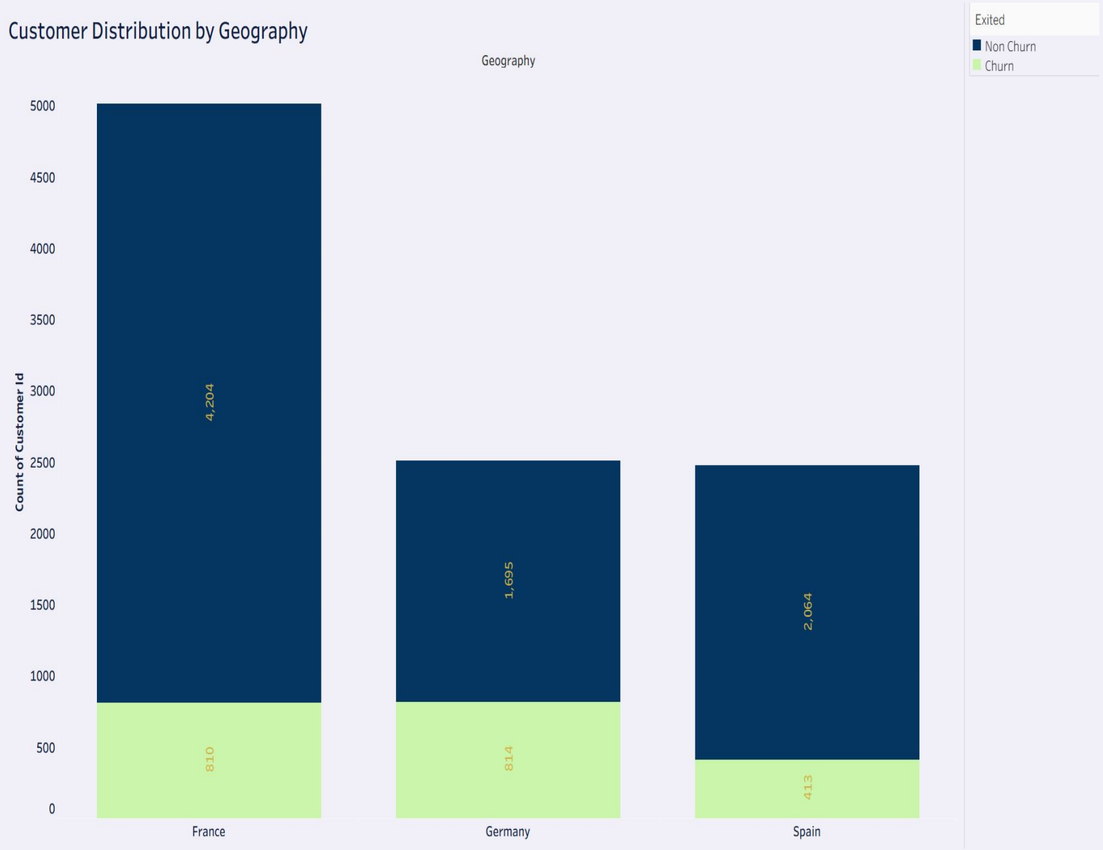
Saldo tinggi ada di semua umur, namun churn dominan di 41–60.

Business Meaning

Kehilangan terjadi pada nasabah mapan secara finansial.

Source : *DEEP Mileston 3 Visualization by Tableau*

Cutomers by Geography



📊 Customer Distribution by Geography

Insight

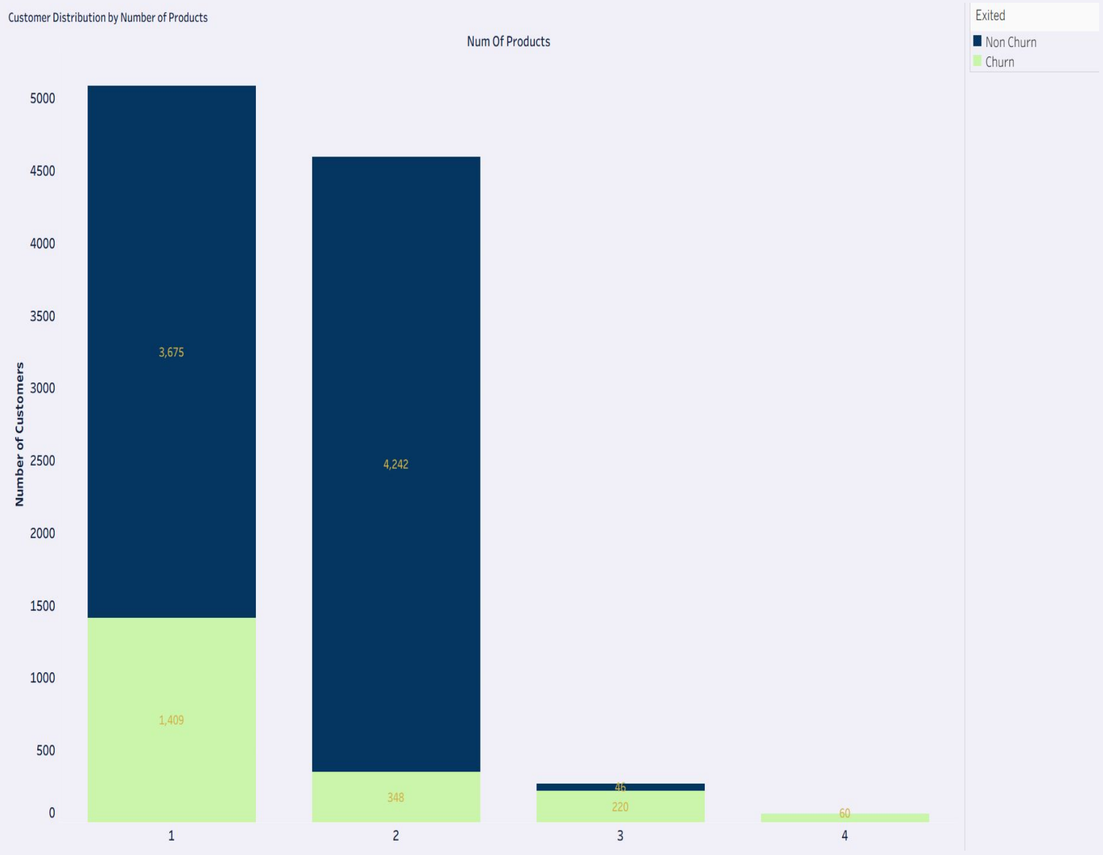
Churn tertinggi di Germany (814) dibanding France (810) & Spain (413).

Business Meaning

Masalah churn terkonsentrasi geografis → perlu strategi spesifik per negara.

Source : *DEEP Mileston 3 Visualization by Tableau*

Customers Distribution by Num of Product



📊 Customer Distribution by Number of Products

Insight

Nasabah dengan 1 produk = 1.409 churn (tertinggi).

Churn turun drastis di 3–4 produk (220 & 60).

Business Meaning

Product adoption adalah pelindung churn terkuat (root cause utama).

Source : *DEEP Mileston 3 Visualization by Tableau*

Horizon Bank